

TABLE 8—TESTS OF MODELS OF PEER EFFECTS USING ESTIMATES FROM TABLE 7

Model	Math				ELA			
	Number of estimates significant in direction the model suggests	Number of estimates insignificant	Number of estimates significant in opposite direction	Consistent, inconsistent, or neither?	Number of estimates significant in direction the model suggests	Number of estimates insignificant	Number of estimates significant in opposite direction	Consistent, inconsistent, or neither?
(1) Weak monotonicity: Effect of evacuees from top quartile is statistically significantly greater than effect from bottom quartile.	4	0	0	Consistent	3	1	0	Consistent
(2) Strong monotonicity: Effect of evacuees from quartile j is statistically significantly greater than effect from quartile $j - 1$.	4	8	0	Consistent	1	11	0	Neither
(3) Invidious comparison: Effect of evacuees from lower quartiles than the incumbent is positive and effect from evacuees in higher quartiles is negative.	0	6	6	Inconsistent	0	8	4	Inconsistent
(4) Ability grouping: Effect of evacuees from own quartile is statistically significantly greater than effect from higher quartiles.	0	2	4	Inconsistent	0	3	3	Inconsistent

Note: See text for the specific tests conducted for each model.

level, his self-confidence and performance may improve, with the opposite case if he is grouped with higher-achieving peers. We reject this model with or without Bonferroni corrections. For example, for math scores, 6 of 12 tests are significant in the opposite direction, while the other 6 tests are statistically insignificant.

Finally, we test whether students benefit from being grouped with other students of the same level. Hence, we test whether the impact of an increase in evacuees from the incumbent student’s same quartile is greater than the impact from higher-achieving evacuees. The focus on comparing the effect of similar peers to higher achievers and not lower achievers allows us to distinguish this model from monotonicity. While grouping is thought to be useful because teachers can adjust their teaching to a more appropriate level for the students, we find no evidence that grouping students at the school-grade level by ability is beneficial for low-achieving students.

Hence, we conclude that the best model for describing our results is a nonlinear model with monotonic peer effects. Moreover, results from the nonlinear models show patterns that are more consistent with peer effects than with alternative stories of the impact of Katrina and Rita evacuees on schools. For example, if the inflow of evacuees reduced school resources (either by increasing class size, reducing expenditures per student, or reducing the quality of teachers), both high-achieving